Jean-Guillaume de Damas

Postdoctoral Researcher in Machine Learning for Numerical Analysis

Paris, France · jean-guillaume.de-damas@enpc.fr · 07 70 70 67 75 · jgdedamas.github.io
0009-0008-6812-7530 · Google Scholar · jgdedamas

Summary

Applied mathematician specializing in randomized linear algebra and model reduction. Research focuses on accelerating numerical simulations through hybrid data-driven and physics-informed methods, with particular expertise in large-scale parametric studies and eigenvalue problems.

Experience

ENPC - CERMICS, Postdoctoral Researcher — Paris, France — May 2026 – present — 1 month

- Working at the Centre d'Enseignement et de Recherche en Mathématiques et Calcul Scientifique on the development of hybrid model reduction methods
- Developing numerical methods for accelerating parametric studies in granular material simulations
- Implementing algorithms in the software SCOPI++

CERFACS, Master Intern - Researcher & Scientific Programmer — Toulouse, France — Apr 2022 – Oct 2022 — 7 months

- Researched randomized SVD preconditioners with applications to data assimilation
- Delivered functional C++ code for the European Weather Forecast Center
- Supervised by Selime Gürol

Education

PhD — **INRIA Paris & Sorbonne Université**, Applied Mathematics — Paris, France — Nov 2022 – Mar 2026

- Thesis: Randomized methods for solving eigenvalue problems and computing low-rank matrix approximations
- Supervised by Laura Grigori within the ERC project EMC2

MSc — **ISAE-SUPAERO**, Engineering — Toulouse, France — Sept 2018 – present

- Machine learning: Algorithms for SVM, Neural Networks, Reinforcement learning and mathematical background
- Statistics: Bayesian inference, Stochastic algorithms, Non-parametric estimation
- High performance computing: Parallelization in C
- Optimisation: Inverse problem, data assimilation, problem regularization, and numerical challenges in inexact arithmetic

Publications

Randomized Implicitly Restarted Arnoldi Method for the Non-Symmetric Eigenvalue Problem — Oct 2025
Vol. 46, No. 4, pp. 2395–2422. ISSN: 1095-7162.
Jean-Guillaume de Damas, Laura Grigori
10.1137/24m1674352 (SIAM Journal on Matrix Analysis and Applications)

Randomized Krylov-Schur Eigensolver with Deflation — Jan 2025
Jean-Guillaume de Damas, Laura Grigori
10.48550/ARXIV.2508.05400 (arXiv)

Randomized Orthogonalization and Krylov Subspace Methods: Principles and Algorithms Jan 2025
Jean-Guillaume de Damas, Laura Grigori, Igor Simunec, Edouard Timsit
10.48550/arXiv.2512.15455 (arXiv)

Selected Talks

Randomized Linear Algebra for the Eigenvalue Problem SIAM Conference on Applied Linear Algebra
Jan 2024

Preconditioning for Data Assimilation with Randomized SVD CERFACS Scientific Seminar
Jan 2022

Teaching

SU Fall 2023
summary

SU

Skills

Languages: Julia, Python, Matlab, R, C, C++

Technologies: PyTorch, Scikit-learn, OpenMP, MPI, Cuda

Expertise: Numerical Analysis, Randomized Linear Algebra, Eigensolvers, Low Rank Approximations

Languages

English: Advanced Level C1 (TOEFL ITP 600/677)

German: Intermediate (A2)

Japanese: Beginner (A1)

Interests

- Piano & Musicology: Learning as a passionate amateur with conservatory-level classes